

# Risk-Aware Shift-Based Decision Trees

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## 1 Introduction

Decision trees such as ID3 and C4.5 are widely used because they are interpretable, efficient, and easy to apply to classification problems [6]. A central part of decision tree learning is the choice of split criterion. Standard criteria such as information gain and gain ratio select attributes by measuring entropy reduction, which reflects how much the class distribution changes after a split [6, 7]. These criteria are effective in many settings, but they primarily optimize average impurity reduction across the resulting branches.

This average-case behavior can hide an important reliability problem. A split may look useful overall while still producing one or more branches whose class distributions remain highly uncertain. In practical domains such as medical diagnosis, this matters because different subsets of the data may contain different levels of ambiguity. A decision path that reaches an uncertain branch may lead to unreliable downstream predictions even when the tree performs well on average.

Prior work has explored alternative split criteria that improve different aspects of reliability. Confidence-based methods, such as the Class Confidence Proportion Decision Tree (CCPDT), replace entropy with measures of class dominance within the partitions created by a split [5]. Other approaches use statistical confidence bounds to determine whether a split is supported by enough evidence, especially in streaming settings [2]. These methods improve split selection in important ways, but they do not directly optimize for the worst branch produced by a split.

To address this limitation, this paper proposes a risk-aware shift criterion for decision tree learning. The shift component measures how much the class distribution changes after a split, while the risk component penalizes the most uncertain branch produced by that split. By combining these terms, the method favors splits that are both informative and reliable in the worst case.

The contributions of this paper are threefold. First, it introduces a split criterion that combines distributional shift with a worst-case branch uncertainty penalty. Second, it implements this criterion in a recursive decision tree learner that supports both numeric and categorical attributes. Third, it evaluates the method against gain ratio and class-confidence baselines using accuracy, average shift, average risk, and tree complexity. The goal is not only to maintain predictive performance, but also to produce decision trees whose selected splits are less likely to contain unreliable branches.

## 2 Related work

Decision tree learning has been widely studied, and much of the classical work focuses on how to choose splits during tree construction. Information gain, used in ID3, selects the attribute that gives the largest reduction in entropy. Gain ratio, used in C4.5, adjusts information gain to reduce the tendency to favor attributes with

many values [7]. Both criteria are based on average impurity reduction across child nodes. As a result, they can prefer a split that is strong in aggregate even if one branch remains ambiguous.

Several works have explored split criteria based on class confidence rather than entropy. Liu et al. propose the Class Confidence Proportion Decision Tree (CCPDT), which evaluates how strongly the resulting partitions are associated with particular classes [5]. This is especially relevant for imbalanced datasets, where entropy-based criteria may not fully capture the usefulness of class-dominant partitions. However, CCPDT aggregates confidence across the split and does not explicitly penalize the single least reliable branch.

De Rosa and Cesa-Bianchi introduce a different form of confidence in decision tree learning for data streams [2]. Their method uses statistical confidence bounds to decide when there is enough evidence to commit to a split. This improves robustness when data arrive over time or when split estimates are uncertain. However, this notion of confidence concerns the reliability of the split estimate, rather than the class uncertainty of the branches after the split has been made.

The proposed method is also related to broader ideas from robust decision-making. Robust optimization emphasizes solutions that remain acceptable under unfavorable conditions, rather than solutions that only perform well on average [1]. This perspective motivates the use of a worst-case risk term in split selection. In the decision tree setting, the unfavorable case is the least reliable branch produced by a candidate split.

Measures of distributional change are related to divergence-based metrics that quantify differences between probability distributions, such as Kullback–Leibler divergence [4]. These measures provide a way to describe how much a child node’s class distribution differs from the parent distribution. The shift component of the proposed method follows this intuition, but uses a simple absolute-difference distance so that the score can be computed directly from class proportions.

Overall, existing split criteria typically emphasize either average distributional change, average class dominance, or statistical confidence in the split estimate. The proposed approach combines distributional shift with an explicit penalty on worst-case branch uncertainty. This makes the method complementary to prior decision tree criteria while focusing on a reliability concern that standard average-based measures can miss.

## 3 Approach

The proposed method modifies the split-selection step of top-down decision tree learning. Like ID3 and C4.5, the algorithm recursively partitions the training data by selecting one split at each internal node and assigning majority-class predictions to leaves [6, 7]. The main difference is that the proposed tree does not select the split

that maximizes information gain or gain ratio. Instead, it evaluates each candidate split using two quantities: a distributional shift term that measures how informative the split is, and a risk term that measures the least reliable branch produced by the split.

Let  $S$  denote the set of training examples that reach the current node, let  $A$  denote a candidate splitting attribute, and let  $S_v$  denote the subset of examples assigned to branch  $v$  after splitting on  $A$ . For a numeric attribute, the candidate split is binary and  $v$  corresponds to the two threshold outcomes. For a categorical attribute, each observed category value defines a branch. The algorithm selects the candidate split with the largest risk-aware shift score:

$$\text{Score}_\beta(S, A) = \text{Shift}(S, A) \cdot \max(0, 1 - \beta \cdot \text{Risk}(S, A))$$

where  $\beta \geq 0$  controls how strongly the method penalizes risky splits. When  $\beta = 0$ , the method ignores risk and ranks splits only by distributional shift. As  $\beta$  increases, splits with uncertain branches receive lower scores. The maximum term prevents the penalty factor from becoming negative when larger  $\beta$  values are tested.

### 3.1 Distributional Shift

The shift component measures how much the class distribution changes after applying a split. A useful split should produce child nodes whose class distributions differ from the parent distribution. This captures the same intuition as entropy-based criteria: a good split should make the children more class-specific than the parent. However, rather than computing entropy reduction, the proposed method directly compares the parent and child class distributions.

Let  $p(c | S)$  denote the class distribution in the parent node and  $p(c | S_v)$  the distribution in each branch  $v$ , and where  $c$  denotes a class label. Shift is defined as the weighted average distance between the parent and child distributions:

$$\text{Shift}(S, A) = \sum_{v \in \text{Values}(A)} \frac{|S_v|}{|S|} D(p(c | S_v), p(c | S))$$

where  $D$  is a distance measure between probability distributions. In this work,  $D$  is the absolute difference between class probabilities:

$$D(p, q) = \sum_c |p(c) - q(c)|$$

The branch weights  $\frac{|S_v|}{|S|}$  prevent small branches from dominating the shift value. A branch containing only a few examples may have a class distribution very different from the parent, but its contribution is limited by its support. This makes the shift term sensitive to meaningful distributional changes while still accounting for the size of each resulting partition.

### 3.2 Worst-Case Branch Risk

The risk component measures the worst-case classification uncertainty among the branches produced by a split. For each branch  $v$ , the dominant class proportion is  $\max_c p(c | S_v)$ . If a leaf were created at that branch, this value would be the confidence of the majority-class prediction. The corresponding branch uncertainty is:

$$1 - \max_c p(c | S_v)$$

The split-level risk is the maximum branch uncertainty:

$$\text{Risk}(S, A) = \max_{v \in \text{Values}(A)} \left( 1 - \max_c p(c | S_v) \right)$$

Thus,  $1 - \text{Risk}(S, A)$  is the minimum class dominance over all branches. This makes the criterion conservative: a split can receive a large shift value only if it changes the class distribution, but it can still be penalized if even one resulting branch remains highly mixed. This differs from average confidence criteria, such as CCPDT, because the weakest branch directly affects the score instead of being hidden inside an average [5]. The design is motivated by the same general concern as robust optimization: decisions should avoid solutions whose worst outcome is unreliable, even when the average behavior appears acceptable [1].

### 3.3 Combined Objective

The multiplicative form of the score forces informativeness and reliability to act together. If a split has no distributional shift, its score is zero regardless of its risk. If a split has high shift but also high worst-case risk, the risk penalty reduces its score. This is important because a purely shift-based criterion may prefer a split that strongly separates one region of the data while leaving another branch ambiguous. Conversely, a purely confidence-based criterion may prefer safe but weak splits that do not substantially change the class distribution. The proposed score is intended to favor splits that satisfy both conditions.

The parameter  $\beta$  controls this trade-off. Lower values of  $\beta$  produce behavior closer to a shift-only criterion, while higher values produce more conservative trees. In the empirical evaluation,  $\beta$  is swept over multiple values so that the effect of risk sensitivity can be compared against gain ratio and class confidence baselines.

### 3.4 Algorithm

The tree-building procedure follows the standard recursive structure of decision tree learning. At each node, the method first checks for stopping conditions. If all examples at the node belong to the same class, the node becomes a leaf. If no valid split remains, the node also becomes a leaf and predicts the majority class of the examples reaching that node.

If splitting is possible, the algorithm evaluates every candidate split. Numeric attributes are sorted, and candidate thresholds are placed between consecutive distinct values. Each threshold produces a binary split. Categorical attributes are evaluated by grouping examples according to their observed category values. For every candidate, the algorithm computes the child class counts, the shift value, the risk value, and the risk-aware score. The best split is then used to create an internal node, and the same procedure is applied recursively to each child.

DecisionTree( $S$ , Attributes,  $\beta$ ):

```
If all examples in  $S$  belong to the same class:
    return Leaf(class)
```

```
If Attributes is empty:
    return Leaf(majority_class( $S$ ))
```

```

For each candidate split produced by Attributes:
  Compute child class distributions
  Compute Shift(S, split)
  Compute Risk(S, split)
  Compute Score_beta(S, split)

Let split_best = candidate with highest Score_beta

Create a decision node using split_best

For each branch v of split_best:
  Let S_v = examples assigned to branch v

  If S_v is empty:
    Attach Leaf(majority_class(S))
  Else:
    Attach DecisionTree(S_v, updated Attributes, beta)

return the decision node

```

In the implementation, numeric attributes remain available for later descendants because they may be split again at different thresholds, while categorical attributes are removed after they have been used. When two candidate splits have effectively equal scores, the implementation uses deterministic tie-breaking. It first prefers the candidate with higher shift, and if the shift values are also equal, it prefers the candidate with lower risk. This preserves the main priority of the proposed score while making the final tree stable across repeated runs.

### 3.5 Evaluation Metrics Produced by the Method

In addition to producing class predictions, each internal node stores the score, shift, and risk of the split selected at that node. These stored values allow the learned tree to be analyzed after training. The empirical evaluation reports classification accuracy to measure predictive performance, average shift to measure how strongly selected splits change class distributions, average risk to measure the typical uncertainty of selected splits, and tree size statistics such as maximum depth and total number of nodes. These measurements make it possible to determine whether the proposed method changes only accuracy, or whether it also changes the reliability and structure of the resulting trees.

## 4 Empirical Evaluation

### 4.1 Evaluation Criteria

The empirical evaluation measures both predictive performance and the structural behavior of the learned trees. Accuracy is used as the primary predictive metric. Since the proposed method is designed to change the quality of selected splits rather than only the final prediction rate, the evaluation also reports average shift and average risk across internal tree nodes. Average shift measures how strongly the selected splits change class distributions, while average risk measures the uncertainty of the branches selected by the tree. Lower average risk indicates that the tree is selecting splits whose branches are more class-dominant on average.

Tree complexity is measured using maximum depth and total number of nodes. These metrics are important because split criteria

can improve accuracy by producing larger or deeper trees. A useful split criterion should therefore be evaluated not only by accuracy, but also by whether it changes the size and reliability of the resulting model.

### 4.2 Experimental Data and Procedures

Experiments were conducted on three binary classification datasets. The Pima Indians Diabetes dataset contains 768 examples with numeric medical features related to diabetes diagnosis [8]. The Breast Cancer Wisconsin Diagnostic dataset contains 569 examples with numeric cell-nuclei measurements and benign/malignant labels [9]. The Heart Disease dataset contains 920 examples with a mix of numeric and categorical clinical attributes [3]. For the heart disease data, the original multiclass diagnosis value was converted into a binary label indicating either no disease or disease present.

The proposed Risk-Aware Shift tree was compared against two baselines. The first baseline uses gain ratio, representing the standard C4.5-style entropy-based split criterion. The second baseline uses class confidence, representing a class-dominance split criterion related to CCPDT. Risk-Aware Shift was evaluated using a beta sweep from  $\beta = 0.0$  to  $\beta = 1.0$ . For direct comparison against the baselines, the results table reports the  $\beta = 0.6$  version, because this value gave the best accuracy on the Pima Indians Diabetes dataset and provided a representative mid-range risk penalty.

Each method was evaluated using repeated stratified 5x2 cross-validation. For each of five repetitions, the examples were stratified by class and divided into two halves; the halves were then swapped to produce two train/test folds. This produced ten folds per dataset. Missing numeric values were imputed using the median value computed from the training portion of each fold, and missing categorical values were assigned a separate missing-value category. This avoids using test-set information during preprocessing.

### 4.3 Results and Analysis

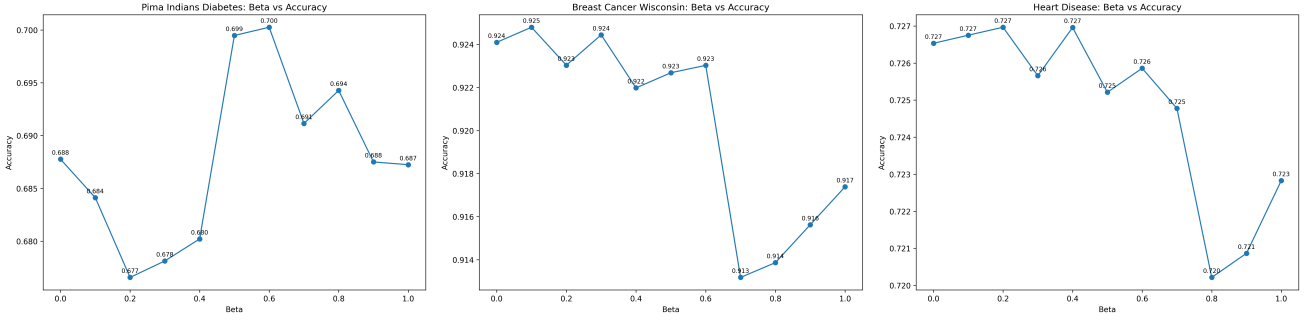
Table 1 shows that the proposed method is competitive with the two baselines while producing substantially higher average shift. At  $\beta = 0.6$ , Risk-Aware Shift achieved higher accuracy than gain ratio on all three datasets: 0.700 versus 0.684 on Pima Indians Diabetes, 0.923 versus 0.916 on Breast Cancer Wisconsin, and 0.726 versus 0.725 on Heart Disease. On the Pima Indians Diabetes dataset, Risk-Aware Shift also achieved the highest accuracy of the three methods, outperforming both gain ratio and class confidence. It produced a much shallower tree than gain ratio, reducing maximum depth from 53.6 to 10.0 and total nodes from 182.0 to 141.2. This suggests that the proposed criterion selected more informative splits without requiring a very deep tree.

On the Breast Cancer Wisconsin dataset, Risk-Aware Shift achieved accuracy similar to class confidence and higher than gain ratio. Class confidence produced the lowest average risk on this dataset, but Risk-Aware Shift produced the highest average shift and the smallest tree among the three methods. This indicates that the proposed criterion selected splits that strongly changed the class distribution while keeping the tree compact.

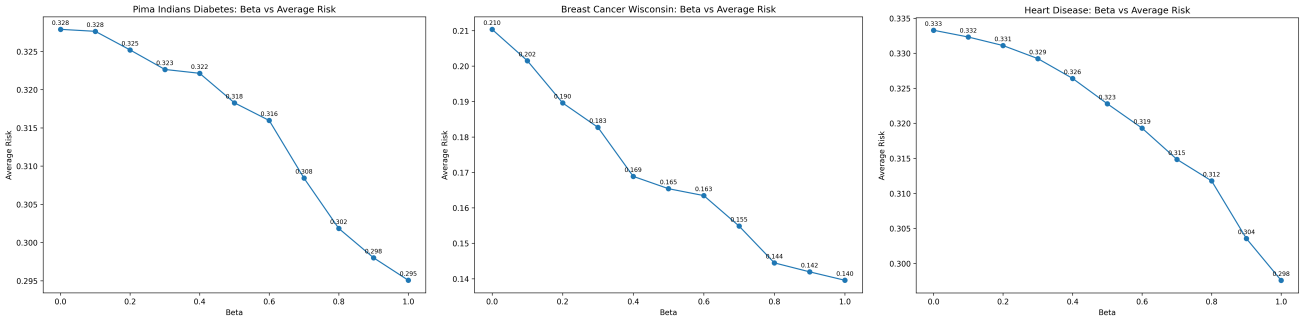
On the Heart Disease dataset, class confidence achieved the highest accuracy and lowest average risk among the three direct-comparison methods. Risk-Aware Shift performed slightly above

**Table 1: Mean results across repeated stratified 5x2 cross-validation. Risk-Aware Shift is shown at  $\beta = 0.6$  for direct comparison with the two baselines.**

| Dataset                 | Method           | Accuracy | Avg. Shift | Avg. Risk | Max Depth | Nodes |
|-------------------------|------------------|----------|------------|-----------|-----------|-------|
| Pima Indians Diabetes   | Gain Ratio       | 0.684    | 0.084      | 0.326     | 53.6      | 182.0 |
| Pima Indians Diabetes   | Class Confidence | 0.687    | 0.194      | 0.273     | 16.3      | 150.4 |
| Pima Indians Diabetes   | Risk-Aware Shift | 0.700    | 0.333      | 0.316     | 10.0      | 141.2 |
| Breast Cancer Wisconsin | Gain Ratio       | 0.916    | 0.201      | 0.159     | 9.2       | 30.0  |
| Breast Cancer Wisconsin | Class Confidence | 0.924    | 0.285      | 0.135     | 7.1       | 24.4  |
| Breast Cancer Wisconsin | Risk-Aware Shift | 0.923    | 0.428      | 0.163     | 5.0       | 22.2  |
| Heart Disease           | Gain Ratio       | 0.725    | 0.170      | 0.321     | 26.3      | 194.9 |
| Heart Disease           | Class Confidence | 0.732    | 0.283      | 0.287     | 13.1      | 181.3 |
| Heart Disease           | Risk-Aware Shift | 0.726    | 0.375      | 0.319     | 9.6       | 178.0 |



**Figure 1: Accuracy of Risk-Aware Shift as the risk penalty parameter  $\beta$  changes.**



**Figure 2: Average risk of Risk-Aware Shift as the risk penalty parameter  $\beta$  changes. Higher values of  $\beta$  generally reduce average risk, showing the intended effect of the penalty term.**

gain ratio in accuracy and produced much higher average shift. It also reduced maximum depth substantially compared with gain ratio, from 26.3 to 9.6. The heart disease results therefore show a trade-off: class confidence was best for risk and accuracy, while Risk-Aware Shift produced more distributionally separated splits and a shallower tree.

Figures 1 and 2 show the effect of changing  $\beta$ . The accuracy curves indicate that the best risk penalty depends on the dataset. Pima Indians Diabetes achieved its highest accuracy at  $\beta = 0.6$ , Breast Cancer Wisconsin peaked at  $\beta = 0.1$ , and Heart Disease peaked at  $\beta = 0.2$ . This suggests that the optimal amount of risk penalization depends on the amount of overlap and noise in the dataset.

The risk curves show a clearer pattern. As  $\beta$  increases, average risk generally decreases across all three datasets. The lowest average risk in the sweep occurs at  $\beta = 1.0$  for each dataset. For example, Pima Indians Diabetes decreases from 0.328 at  $\beta = 0.0$  to 0.295 at  $\beta = 1.0$ , Breast Cancer Wisconsin decreases from 0.210 to 0.140, and Heart Disease decreases from 0.333 to 0.298. This supports the intended behavior of the proposed criterion: increasing the penalty on worst-case branch uncertainty leads the tree to select lower-risk splits.

Overall, the results support the hypothesis that Risk-Aware Shift can maintain competitive accuracy while changing the kinds of splits selected by the tree. The method consistently increases average shift compared with gain ratio and class confidence, and the

beta sweep demonstrates that the risk penalty can reduce average risk when stronger reliability is desired. The main trade-off is that the beta value must be chosen carefully, since the lowest-risk model is not always the most accurate model.

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## 5 Conclusion

This paper presented Risk-Aware Shift, a modified decision tree split criterion that combines distributional shift with a penalty for worst-case branch uncertainty. The method is designed to address a limitation of standard split criteria: a split can appear strong on average while still producing at least one unreliable branch. By incorporating both informativeness and branch-level reliability into the split score, the proposed criterion encourages trees to select splits that meaningfully change the class distribution without ignoring uncertain outcomes.

The empirical results show that Risk-Aware Shift is competitive with gain ratio and class confidence baselines across three medical classification datasets. At  $\beta = 0.6$ , the proposed method achieved higher accuracy than gain ratio on every dataset tested, while also producing consistently higher average shift. This indicates that it selected splits with stronger distributional separation without sacrificing predictive performance relative to the standard entropy-based baseline. It also produced shallower or smaller trees than gain ratio in the reported experiments, suggesting that the criterion can find useful splits without relying on excessive tree depth. The beta sweep further showed that increasing the risk penalty generally reduced average risk, confirming that the parameter controls the intended reliability trade-off.

The results also show that the best beta value depends on the dataset. The lowest-risk setting was not always the most accurate setting, so the method should be treated as a tunable criterion rather than a fixed replacement for existing split rules. Future work could study automatic beta selection, evaluate the method on larger and more diverse datasets, and explore alternative distance functions for the shift term. Overall, Risk-Aware Shift provides a simple way to incorporate worst-case branch reliability into decision tree learning while preserving the interpretability and recursive structure of standard decision trees.

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